

A survey of architecture description languages

PC Clements - Proceedings of the 8th international workshop ..., 1996 - [ieeexplore.ieee.org](#)

... In our **survey**, the domain consisted of the set of languages that might be considered ADLs;

"... of a **survey** questionnaire; the questionnaire is the manifestation of a **framework** of important ...

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Investigating architecture description languages (adls) a systematic literature review

S Hussain - 2013 - [diva-portal.org](#)

... That is then when the need for **Architecture Description Language** arose; when systems

became ... **framework** and motivation of the topic. Its main purpose is to justify why the **review** is ...

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Architecture Description Language–Systematic Literature Review

TE Babeker, H Ammar - 2021 - [scholar.archive.org](#)

... **Overview**, "Technical and Historical **Overview** of MetaH, " 2000. [40] JH McDuffie, "Using

the **architecture description language** MetaH for designing and prototyping an embedded ...

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A study of architecture description languages from a model-based perspective

W Qin, S Malik - 2005 Sixth International Workshop on ..., 2005 - [ieeexplore.ieee.org](#)

... **state** machine (OSM) model and briefly discusses the Mescal **Architecture Description**

Language (... BABEL [24] is an ADL for retargeting the popular processor simulation **framework** ...

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A classification and comparison framework for software architecture description languages

N Medvidovic, RN Taylor - IEEE Transactions on software ..., 2000 - [ieeexplore.ieee.org](#)

... While we believe that this **taxonomy** gives the architect a sound foundation for selecting

an ADL and orients discourse toward problem solving, it is still very much a preliminary ...

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Architecture description languages

P Mishra, N Dutt - Customizable Embedded Processors, 2007 - Elsevier

... The phrase "**architecture description language**" has been ... This section presents the **survey**

using content-based classi.... retargetable compiler and simulator generation **framework** [11]. An ...

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Embedded architecture description language

J Li, NT Pilkington, F Xie, Q Liu - Journal of Systems and Software, 2010 - Elsevier

In the **state-of-the-art** hardware/software (HW/SW) co-design of embedded systems, there is

a lack of sufficient support for architectural specifications across HW/SW boundaries. Such ...

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Introduction to Architecture Description Languages

P Mishra, N Dutt - Processor Description Languages, 2008 - Elsevier

... The phrase "**architecture description language**" is used in the ... Therefore, the ideal "**Architecture**

Description Language" (... Chapter 7 presents ASIP Meister **framework**. This chapter ...

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Architecture: Description Languages.

J Ryoo - 2010 - books.google.com

... a semiautomated language **framework** that facilitates visual and ... definition of the meta-**taxonomy** of architectural views with a ... **framework** for software **architecture description language**. ...

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The architecture description language model

P Klein - Graph Transformations and Model-Driven Engineering ..., 2010 - Springer

... not specifically encourage architects to design a robust **framework** structure for the PiS phase.

... Data abstraction modules encapsulate **state** or a template for **state**; they hide the internal ...

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A systematic literature **review** on the description of software architectures for systems of systems

M Guessi, VVG Neto, T Bianchi, KR Felizardo... - Proceedings of the 30th ..., 2015 - dl.acm.org

... In addition, we also found a reference **framework** for realizing Component-Based Systems of ... bound to a specific ADL (**Architecture Description Language**) or implementation language. ...

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Are we there yet? Analyzing architecture description languages for formal analysis, usability, and realizability

M Ozkaya, C Kloukinas - 2013 39th Euromicro Conference on ..., 2013 - ieeexplore.ieee.org

... that current ADLs provide for them, our **survey** focuses on what we believe keeps industry ... Middleware-layer connector synthesis: Beyond **state** of the **art** in middleware interoperability...

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Architecture Description Language (ADL): a survey

Z Salamaty, K Nikfarjam, M Mehmandoust - 2016 - sid.ir

... : A new **architecture description language** called SADL, intended for the expression of software architecture hierarchies that are to be analyzed formally. The entire SADL **framework** is ...

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The ArchC architecture description language and tools

R Azevedo, S Rigo, M Bartholomeu, G Araujo... - International Journal of ..., 2005 - Springer

... This paper presents an **architecture description language** (ADL) called ArchC, which is an open-... (6) used LISA to create a **framework** for hardware/software co-simulation, by connecting ...

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Bridging the gap between IEEE 1471, an **architecture description language**, and UML

MM Kandé, V Crettaz, A Strohmeier... - Software and Systems ..., 2002 - Springer

... architecture, which instantiates the conceptual **framework** defined in ANSI/IEEEStd-1471 ... **state** machines. At this stage, we use the protocol **state** machines instead of standard UML **state** ...

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Architectural description languages (adls) vs uml: a **review**

RK Pandey - ACM SIGSOFT Software Engineering Notes, 2010 - dl.acm.org

... These notations usually provide both a conceptual **framework** and a concrete syntax for characterizing software architectures. They also provide tools for parsing, displaying, compiling, ...

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SMADL—The social machines **architecture description language**

LM Nascimento - 2014 - repositorio.ufpe.br

... Hence, the goal of this investigation is to identify, evaluate, and synthesize **state-of-the-art** domain-specific programming practices, and then gather evidence of what has been achieved ...

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A comprehensive approach for the development of modular software architecture description languages

EM Dashofy, A Hoek, RN Taylor - ACM Transactions on Software ..., 2005 - dl.acm.org

... tion of an **architecture description language** into modules, ... Additionally, we will provide a brief **overview** at the end of the ... been on the creation of the **framework**, we intend to use this ...

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Designing software architectures with an aspect-oriented **architecture description language**

J Pérez, N Ali, JA Carsí, I Ramos - ... CBSE 2006, Västerås, Sweden, June 29 ..., 2006 - Springer
... Aspect-Oriented **Architecture Description Language** (AOADL). ... Section 3 gives an **overview**
of the PRISMA approach and ... Despite the fact that .NET **framework** does not provide support ...

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Architecture description languages for programmable embedded systems

P Mishra, N Dutt - IEE proceedings-computers and digital techniques, 2005 - IET

... This paper presents a comprehensive **survey** of existing ADLs and ... The term '**architecture description language**' (ADL) has been used in ... In this paper, we **survey** contemporary ADLs. ...

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An Overview of Software Architecture Description Language and Evaluation Method

Y Lingling, Z Wei - Proceedings of the 2012 International Conference on ..., 2013 - Springer

... , and the unify the existing various ADL in this **framework**. ... Rapide, a kind of **architecture description language** (ADL in ... defines it as a development **framework** that supports component-...

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Architecture Description Languages for Retargetable Compilation.

W Qin, S Malik - 2002 - scholar.archive.org

... It is the primitive form of an **architecture description language**... In this chapter, we will start with a **survey** of the existing ADLs... It is not intended to be a general **framework** for calling ...

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Towards a service-based architecture description language

MW Rennie, VB Misic - 2004 - Citeseer

... In this paper, we **review** the basic operation of software ... the use of a suitable **architecture description language** (ADL). Upon ... A classification and comparison **framework** for software ...

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What industry needs from architectural languages: A survey

I Malavolta, P Lago, H Muccini... - IEEE Transactions ..., 2012 - ieeexplore.ieee.org

... We challenged the **survey** participants on the features elicited from the **state-of-the-art** ALs (... -based languages are grouped into **architecture description language** we call AL. We do not ...

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A brief survey of software architecture

R Land - Mälardalen Real-Time Research Center (MRTC) ..., 2002 - Citeseer

... probably be considered as a separate **architecture description language** altogether" [12]. ... I will find more thoroughly and keep up with the **state-of-the-art**; I will study the field of CBSE, ...

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A secure software architecture description language

J Ren, R Taylor - Workshop on Software Security Assurance Tools ..., 2005 - researchgate.net

... We first give an **overview** of our existing architectural description language, and then we outline the ... The XACML **framework**, combined with our explicit modeling of architectural context, ...

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A family of languages for architecture description

M Voelter - 8th OOPSLA Workshop on Domain-Specific Modeling ..., 2008 - Citeseer

... Just using an existing, generic **architecture description language** (such as UML or one of the ... a **statemachine** is probably relatively straight forward. Here is one way of integrating **state** ...

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Architectural description of embedded systems: a systematic review

M Guessi, EY Nakagawa, F Oquendo... - Proceedings of the 3rd ..., 2012 - dl.acm.org

... Thus, the main contribution of this paper is to present and discuss results of a systematic **review**, aiming at providing this wide and, at the same time, deep panorama. We found out that ...

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Systematic language extension mechanisms for the MontiArc **architecture description language**

A Butting, A Haber, L Hermerschmidt, O Kautz... - ... and Applications: 13th ..., 2017 - Springer

... Based on **state-of-the-art** software language composition mechanisms [4], we have conceived the MontiArc ADL [17] for extensive systematic extension on top of the MontiCore [20] ...

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ArchC: A SystemC-based **architecture description language**

S Rigo, G Araujo, M Bartholomeu... - 16th Symposium on ..., 2004 - ieeexplore.ieee.org

This paper presents an **architecture description language** (ADL) called ArchC, which is an open-source SystemC-based language that is specialized for processor architecture ...

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Architecture Description Languages for Automotive Systems—A Literature Review

S Björnander, L Grunske - 2008 - es.mdh.se

... MetaH was extended into the Avionics **Architecture Description Language** (AADL). The name ... The NFP Modelling **Framework** is a set of extensions to specify non-functional annotations. ...

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DAOP-ADL: an architecture description language for dynamic component and aspect-based development

M Pinto, L Fuentes, JM Troya - International Conference on Generative ..., 2003 - Springer

... DAOP-ADL: An **Architecture Description Language** ... Our approach is a sort of middleware AO **framework**. DAOP differs ... The unmarshall aspect retrieves the persistent **state** of the ...

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General purpose ADL (architecture description language)

A Khara - 2014 - web4.cs.columbia.edu

In the design of system architecture¹, a blueprint of the system typically comes second, after requirements are gathered in the process of architecture development. This is followed by ...

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Architecture description language (ADL)-driven software toolkit generation for architectural exploration of programmable SOC

P Mishra, A Shrivastava, N Dutt - ACM Transactions on Design ..., 2004 - dl.acm.org

... 2.1 Brief **Survey** of ADLs ... all the alternatives without using an automated exploration **framework**. This article presented an **architecture description language** (ADL)-Driven exploration ...

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Defining an Architecture Description Language for Dassault Systèmes

Y Ledru, R Sanlaville, J Estublier - 4th International Software ..., 2000 - academia.edu

... It also corresponds to **state** of the **art** technology in the field of open architectures. ... This paper has presented our first efforts in the definition of an **Architecture Description Language** for a ...

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Designing an architecture description language for nontechnical actors and purposes (NOTE-ADL)

S Strohmeier, F Gross - Journal of Enterprise Information ..., 2020 - emerald.com

... In section 2, we give a short literature **overview** about the ... of an ADL that serves as a **framework** for our conceptual model. ... Following up on the illustration of the current research **state** ...

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Architecture description languages taxonomies **review**: a special focus on self-adaptive distributed embedded systems

F Boutekkouk - International Journal of Technology Diffusion (IJTD), 2021 - igi-global.com

... To do this, a search methodology was followed enabling a systematic **review**. Results showed that only a few standard ADL have been accepted by the embedded industry favoring ...

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Teaching computer architecture using an architecture description language

S Rigo, M Juliato, R Azevedo, G Araújo... - Proceedings of the ..., 2004 - dl.acm.org

... They have the opportunity of learning details about architectures that are **state-of-the-art** in the industry, which is an extra motivation for the course. Moreover, students are getting some ...

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A formal concurrency model based **architecture description language** for synthesis of software development tools

W Qin, S Rajagopalan, S Malik - ... of the 2004 ACM SIGPLAN/SIGBED ..., 2004 - dl.acm.org

... In order to advance the **state** along an edge in the **state** diagram, the **state** machine must ...

They are new to this **framework** and are part of the new contributions of the paper. All control...

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A model for specifying rationale using an **architecture description language**

L De Silva, D Balasubramaniam - European Conference on Software ..., 2011 - Springer

... Building architecture models of existing software and comparing system evolution with

and without the Grasp **framework** will enable us to carry out this evaluation. Grasp is currently ...

☆ Save  Cite Cited by 11 Related articles All 7 versions

Survey on the style and description of software architecture

C Hai-Shan - ... on Computer Supported Cooperative Work in ..., 2004 - ieeexplore.ieee.org

... **Architecture Description Language** (ADL) provides the methods and tools to describe the software architecture. ADL is the major issue in the research of software architecture. ADL not ...

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MiADL: An **Architecture Description Language** for design Space Exploration

J Metrôlho, CA Silva, C Couto, A Tavares - IFAC Proceedings Volumes, 2006 - Elsevier

... The operation layer captures the flow of operations in the hardware using finite **state** machines.

MADL allows modeling a larger variety of machines, when compared to LISA. The ArchC ...

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Inheritance in architecture description languages

T Opluštil - Proceedings of the Week of Doctoral Students ..., 2003 - Citeseer

... the Java language in mind as the primary implementation language of the SOFA **framework**.

... On the other hand, we will **state** arguments favoring the specification reuse purpose of the ...

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A research landscape on formal verification of software architecture descriptions

C Araujo, E Cavalcante, T Batista, M Oliveira... - IEEE ..., 2019 - ieeexplore.ieee.org

... This work contributes with (i) a structured **overview** and **taxonomy** of the current **state** of the

art ... Her current research interests include **architecture description language**, cloud computing, ...

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A highly-extensible, XML-based **architecture description language**

EM Dashofy, A Van der Hoek... - ... Working IEEE/IFIP ..., 2001 - ieeexplore.ieee.org

... To support this, we will need an ADL that can store the **state** of components and connectors,

events that are traversing the architecture, and a method to represent an architecture that is ...

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Enterprise architecture analysis using an **architecture description language**

S Sircar, A Kott - B PJOSBBL, 2000 - apps.dtic.mil

... distributed control **framework**, this scheme also has possibilities for differences in **state**

estimation... : first, superposed over a closed-loop **framework**, the use of model checking techniques ...

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xSADL: an **architecture description language** to specify component-based systems

S Zhang, S Goddard - ... and Computing (ITCC'05)-Volume II, 2005 - ieeexplore.ieee.org

... With the help of the component **framework**, xSADL designs can be transformed into run ...

state machines. The LTS specifications can be given in a process algebra notation, Finite **State** ...

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Software architecture description and UML

P Avgeriou, N Guelfi, N Medvidovic - International Conference on the ..., 2004 - Springer

... with a brief synopsis of the **state-of-the-art** and future trends. ... was to devise an **architecture**

description language, based on ... model in terms of a **state** machine that implements the ...

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π -ADL: an **architecture description language** based on the higher-order typed π -calculus for specifying dynamic and mobile software architectures

F Oquendo - ACM SIGSOFT Software Engineering Notes, 2004 - dl.acm.org

... Enabling specification of dynamic and mobile architectures is a large challenge for an

Architecture Description Language (ADL). This article describes π -ADL, a novel ADL that has ...

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Towards an **architecture description language** for hybrid quantum-classical systems

J Zhao - 2024 IEEE International Conference on Quantum ..., 2024 - ieeexplore.ieee.org

... We aim to provide a robust **framework** for coherently describing, integrating, and ... a **review** of existing ADLs and quantum computing fundamentals and developing a new ADL **framework**...

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